Algebra 1 Unit 8 Assessment Guide

General Information

Unit 8 builds on students' understanding of factoring quadratics from Unit 7. Students factor a quadratic expression or equation to rewrite it in factored form. Students apply their experience of using algebra tiles and their understanding of perfect square trinomials to learn how to rewrite a quadratic expression or equation in vertex form. Students also revisit concepts from Unit 2 when they rewrite a factored form or a vertex form of a quadratic to standard form. A student's fluency of these forms helps support their learning and their understanding of the key features of quadratics.

Teachers may find that the language of the benchmarks in this unit are very closely related. Here are some overlaps.

- MA.912.AR.3.6 and MA.912.AR.3.8 – Students are interpreting the vertex and/or zeros in AR.3.6 and any key feature, which includes the vertex and zeros, in AR.3.8. MA.912.AR.3.6 states that an expression or an equation of a quadratic function is given and the draft item specifications require that items for MA.912.AR.3.8 include equations or functions. For this reason, the guide shows both benchmarks as being tagged when students are interpreting the vertex and/or the zeros for a quadratic given in a real-world context.
- MA.912.AR.3.8 and MA.912.AR.9.6 – Students determine constraints for both benchmarks. AR.3.8 is specifically stated as a domain constraint whereas AR.9.6 does not restrict the constraint to the domain. In most real-world contexts, the constraint should be applied to the domain. Also, AR.9.6 states that students should also determine if solutions are viable or not viable.

This table may help teachers distinguish between the benchmarks.

Benchmark	Benchmark Language Notes	Draft Item Specifications Notes
MA.912.AR.3.6	• An expression or equation should be given. • Only key features students find and/or interpret are the vertex and/or the zeros.	Items require a real-world context.
MA.912.AR.3.7	• A table, equation, or written description should be given. • Students graph the quadratic and/or determine the key features.	Items require a mathematical context.
MA.912.AR.3.8	• "Solve and graph" means students can solve, graph, or both. • Students find and/or interpret all key features. • Students determine domain constraints based on the real-world context.	• Items require a real-world context if students are graphing and/or interpreting key features. • Equations of functions must be given. • A mathematical or a real-world context can be used when the

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<tr><td></td><td></td><td>student is asked to solve the quadratic.</td></tr>
<tr><td>MA.912.AR.9.6</td><td>Students represent constraints.
Students interpret solutions as viable or non-viable.</td><td>Items require a real-world context.</td></tr>
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Often the only real-world quadratic students are exposed to is projectile motion. The real-world quadratics on this assessment and in the curriculum are based on real-world data that is modeled by a quadratic. The equations are often modified to be a better match for the Algebra 1 student, but the contexts are real. Exposing students to a variety of real-world contexts gives the student a richer understanding of where mathematics is used. If your students are struggling with the interpretations of the key features, then helping students build that understanding before starting Unit 9 or consecutively with Unit 9 will support student understanding of the benchmarks in Unit 9 as well as the interpretations of key features of other functions within the course.
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<tr><td>Calculator Information</td><td>The assessment is calculator neutral. If students are allowed a calculator, then teachers may wish to guide students to the number of place values to use for decimal equivalents.</td></tr>
<tr><td>Total Recommended Points</td><td>The total recommended points in this guide is 100 points.</td></tr>
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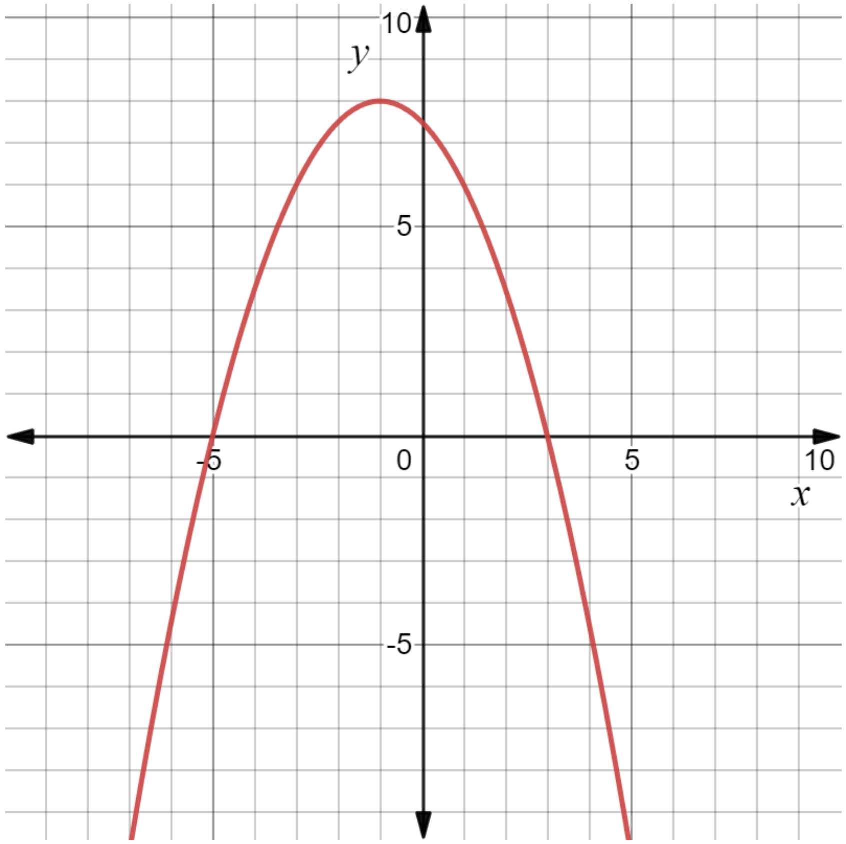
Question	1	Benchmark(s)	MA.912.AR.3.7
	**Recommended Scoring** **(4 Total Points)**	Although the given equation is in factored form, students are not obligated to use this form. Teachers are encouraged to note how students approached graphing the equation. If students use the factored form, then they may have used the intercepts to graph the function. Teachers should look for a graph that is precise, but the graph may not contain all points so the curve may not be as smooth as when a student uses several points to graph.	
Question	**2**	**Benchmark(s)**	MA.912.AR.1.2
$y = -0.5x^2 - x + 7.5$	**Recommended Scoring** **(4 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.	

Question	3	Benchmark(s)	MA.912.AR.3.8

Name	Key Feature	Name	Key Feature
x-intercepts	$(-1, 0)$ and $(3, 0)$	increasing	$x < 1$
y-intercept	$(0, 3)$	decreasing	$x > 1$
vertex	$(1, 4)$	positive	$-1 < x < 3$
axis of symmetry	$x = 1$	negative	$x < -1$ and $x > 3$
domain	R or $-\infty < x < \infty$ or $\{x \mid -\infty < x < \infty\}$	range	$y < 4$ or $\{y \mid y < 4\}$

Recommended Scoring (40 Total Points)	Consider distributing the points so that each key feature is 4 points.

Question	4	Benchmark(s)	MA.912.AR.1.2
$y = 4(x+5)^2 - 11$		Recommended Scoring (4 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	5	Benchmark(s)	MA.912.AR.1.2
$y = (x+2)(x-8)$		Recommended Scoring (4 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	6	Benchmark(s)	MA.912.AR.3.7
Complete each statement. As $x \to -\infty$, $y \to \infty$. As $x \to \infty$, $y \to \infty$.		Recommended Scoring (8 Total Points)	Consider not giving partial credit since students should understand that both ends of a quadratic function have the same end behavior.
Question	7	Benchmark(s)	MA.912.AR.3.6, MA.912.AR.3.8
2016		Recommended Scoring (4 Total Points)	Consider giving students who answer "3 years" partial credit. Students who answer "3 years after 2013" or "2016" are correctly interpreting the context.
Question	8	Benchmark(s)	MA.912.AR.3.7
2015 to 2019		Recommended Scoring (4 Total Points)	Consider partial credit for students who state that the average gas mileage is decreasing for all values greater than (or equal to) 5. This is where the function is decreasing but the context gives the domain as 0 to 9 (2010 to 2019) so the correct interpretation is for the years 2015 to 2019. Students who

			answer 5 to 9 are showing some understanding of the domain but are not correctly stating the answer to the question. Students may also give the years as 5 to 14.574 (or 2005 to 2014-2015 or sometime in 2014) which is interpreting that a negative y-value would not make sense. Although this shows some understanding, the student is not correctly interpretating the context so it is suggested that these students also get some partial credit.
Question	**9a**	**Benchmark(s)**	MA.912.AR.3.6, MA.912.AR.3.8
$(-1, 0)$ and $(14, 0)$		**Recommended Scoring** **(4 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

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<tr><th>Question</th><th>9b</th><th>Benchmark(s)</th><th>MA.912.AR.3.6, MA.912.AR.3.8</th></tr>
<tr><td colspan="2">The year in which there was no relative change of the undernourished population as a percentage of the small countries in the Pacific Ocean is year 2015.

The x-intercepts in this equation are -1 year and 14 years after the year 2001, or years 2000 and 2015, that the relative change was zero. But -1 years after 2001, or year 2000, does not lie in the domain so it doesn’t make sense in the context.</td><td>Recommended Scoring

(8 Total Points)</td><td>Students should notice that $(-1, 0)$ does not make since within the context. Consider giving partial credit if the student correctly interprets that in the year 2000 there is no relative change, but does not note that this value is outside the given domain.

Students may struggle stating the y-value since relative change is not something they have experience with. It is valuable for students to be assessed using contexts they have not seen. If you notice your students struggle with this interpretation, then it is suggested that it is used as a teaching moment before moving on to Unit 9.</td></tr>
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Question	10a-c	Benchmark(s)	MA.912.AR.3.8

	Coordinate	Key Feature (write none if it is not a key feature)	Does the coordinate make sense for the context?	Justification
a.	$(-7.565, 0)$	x-intercept	☐ No ☐ Yes	The domain is 0 to 9. The coordinate is not in the domain.
b.	$(0, 37)$	y-intercept	☐ No ☐ Yes	This is the gas mileage for vehicles sold in Sweden in the year 2010.
c.	$(6, 46)$	vertex	☐ No ☐ Yes	6 years after 2010, or year 2016, the average gas mileage for vehicles sold in Sweden is 46 mpg.

Recommended Scoring **(12 Total Points)**	For each given coordinate, consider partial credit by giving 2 points for the correct name of the key feature and 2 points for the correct justification, which includes stating whether the coordinate makes sense in context.

Question	10d	Benchmark(s)	MA.912.AR.3.8, MA.912.AR.9.6
$0 \leq x \leq 9$ and $37 \leq y \leq 46$		**Recommended Scoring** **(4 Total Points)**	Consider partial credit by giving 2 points for the correct constraints for the x-values and 2 points for the correct constraints for the y-values.